DATA SPACE

# WHAT IS EXPLORATORY DATA ANALYSIS ?

EXPLORATORY DATA ANALYSIS IN POWER BI

Presented by :

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# PHASE ONE

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# FIVE STEPS TO AI COMPS BEFORE EDA

1. understanding business problematic
2. Data collection
3. investigation
4. data analysis
5. the report

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# WHAT IS BUSINESS PROBLEMATIC ?

what the company is trying to solve? , why are we doing this project ? , what is the main issue? , you need to find the business problematic in your data

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# WHAT IS DATA COLLECTION ?

gathering or importing the necessary data to understand the problem, and for the best decision making , you need always to check if the data is sufficient to explain the problematic

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# WHAT IS INVESTIGATION ?

the way you are treating your data (data cleaning , finding outliers , missing values .... )

what is the important skill in this step ?

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# EXAMPLE :

| Satisfaction_Score | Experience_Years | Region | Monthly_Sales | Age | Employee_ID |
|---|---|---|---|---|---|
| 4.1 | 2 | East | 5400 | 26 | E001 |
| 3.8 | 3 | West | 5800 | 29 | E002 |
| 4.5 | 5 | North | 200- | 31 | E003 |
| 88 | 2 | Sout | 6200 | 27 | E004 |
| 4.0 | 12 | East | 7200 | 40 | E005 |
| 3.6 | 1 | West | 99999 | 23 | E006 |
| 4.2 | 2 | east | 6100 | 27 | E007 |
| 4.2 | 2 | East | 6100 | 27 | E007 |
| 4.0 | 4 | North | 5900 | ? | E008 |

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# WHAT IS DATA ANALYSIS ?

the best way to represent the data it means the visualisation using the dashboard,charts,and graphs to summarize insights

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# WHAT IS REPORT ?

A report is a clear and organized document that explains information, summarizes results, or describes something that happened, so that the reader can understand it quickly

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# PHASE TWO

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# WHAT IS EXPLORATORY DATA ANALYSIS?

"An approach of analyzing data sets to summarize their main characteristics, often using statistical graphics and other data visualization methods."

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# SIX STEPS TO EDA

1. Understanding the data structure
2. Identifying missing data
3. Describing the data with descriptive statistics & distributions
4. Identifying outliers
5. Examining and quantifying relationships between variables
6. Forming hypothesis

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## 1 Understanding the data structure

### Continuous

Numerical variables often able to take an infinite set of values.

- Number of stars in space
- Click-through rates
- Distance between two cities

### Categorical

Non-numerical variables, usually text, with two or more group.

- House types
- Country
- Company

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## 2 Identifying missing data

Missing data = when some values are absent (blank or null)

Data is missing with no systematic pattern - the missingness is unrelated to the value itself or other variables.

Example: Survey responses lost due to technical errors

| CITY | Rainfall (inches) | | | |
|---|---|---|---|---|
| | 2.03 | 1.13 | 0.52 | 4.59 |
| SEATTLE | 4.67 | | 2.09 | 4.53 |
| | 0.42 | 2.60 | 1.90 | |
| | 1.35 | 3.40 | 3.75 | 1.75 |
| NYC | | 3.93 | 0.07 | 3.14 |
| | 3.96 | 3.95 | | 3.60 |
| | 4.72 | | 2.27 | 2.68 |
| PARIS | 2.33 | 2.07 | 1.06 | 1.38 |
| | | 4.29 | 4.29 | 1.47 |

Data is missing in a systematic way - there's a relationship between the missingness and other variables or the value itself.

Example: High earners refusing to report income

| CITY | Rainfall (inches) | | | |
|---|---|---|---|---|
| | | | | |
| SEATTLE | 4.67 | 1.75 | 2.09 | 4.53 |
| | 0.42 | 2.60 | 1.90 | 3.14 |
| | 1.35 | 3.40 | 3.75 | 1.75 |
| NYC | 2.68 | 3.93 | 0.07 | 3.14 |
| | 3.96 | 3.95 | 0.52 | 3.60 |
| | 4.72 | 4.72 | 2.27 | 2.68 |
| PARIS | 2.33 | 2.07 | 1.06 | 1.38 |
| | 2.07 | 4.29 | 4.29 | 1.47 |

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## 2 Addressing missing data

Once you've identified missing data, you need to decide how to handle it. The strategy depends on the amount and pattern of missing data.

### Deletion

Remove rows or columns with missing values (use when missingness is small and random)

### Imputation

Fill missing values with mean, median, mode, or predicted values

### Keep as Missing

Treat missing as a separate category (useful when missingness is meaningful)

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## 3 Describing the data

- Minimum
- Maximum
- Mean: sum of all values divided by the number of observations
- Median: the value in the center of a range of values
- Standard Deviation: average amount of difference from the mean of a variable observed across all data points

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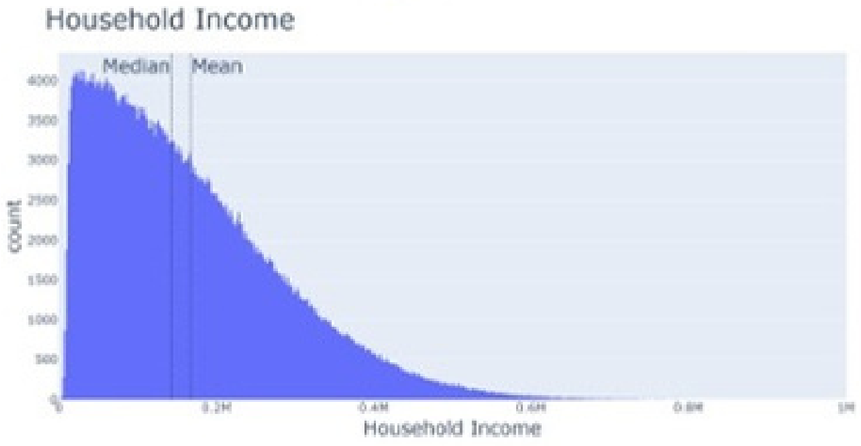
# 3 Describe the data with distributions.




- Median < Mean
- "Right-skewed": the tail is to the right

- Median > Mean
- "Left-skewed": the tail is to the left

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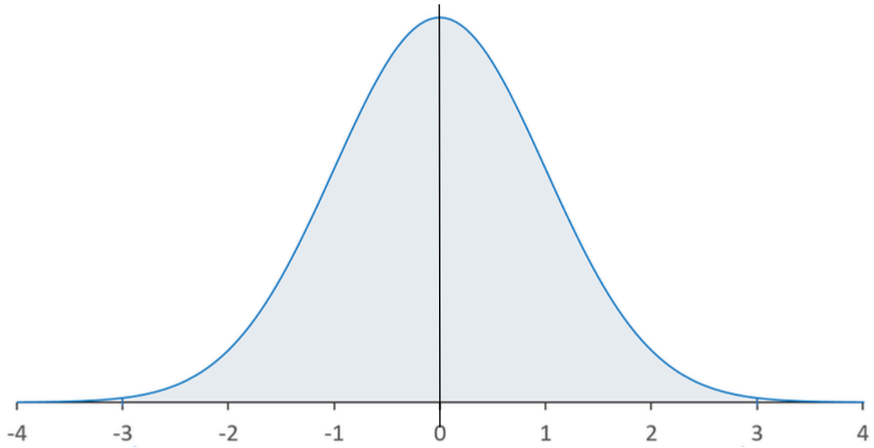
# 3 Describe the data with distributions.

## Normal distributions

-Mean = Median

-A symmetrical, bell-shaped curve where most values cluster around the center.

-Common in nature: heights, test scores, measurement, errors




Mean = Median

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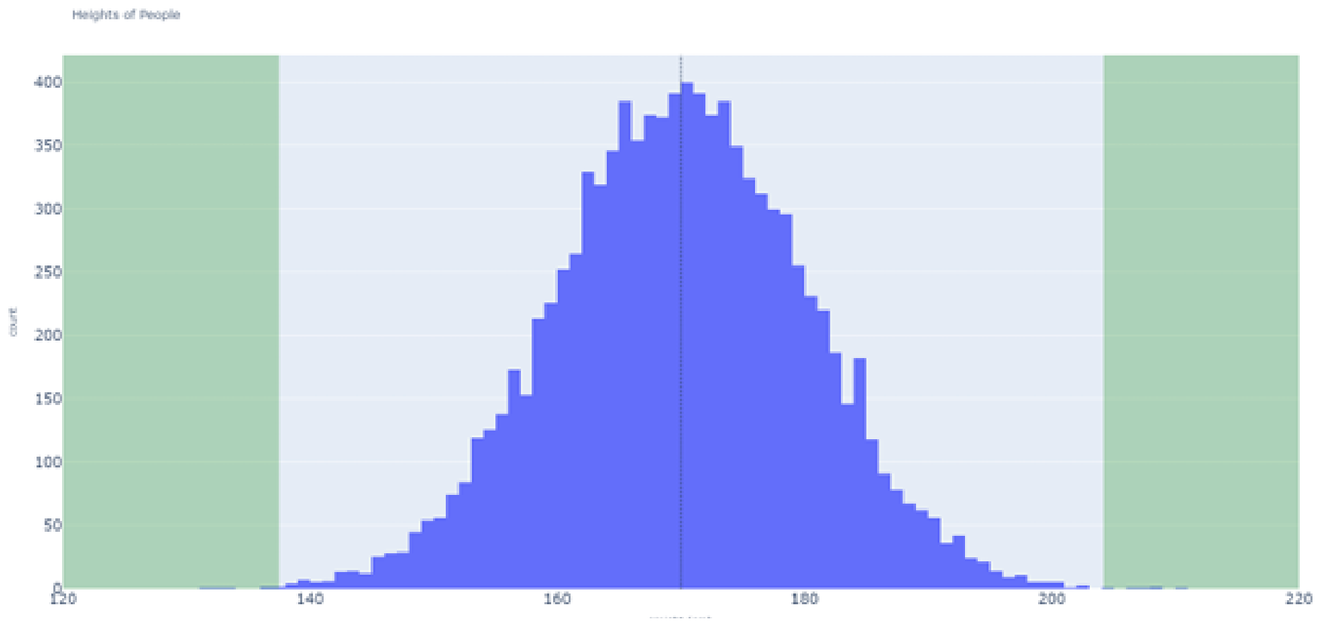
# 4 Identifying outliers

## What is an outlier?

An outlier is a value that is very different from all the other values

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# 4 Finding outliers

Using standard deviation

$$lower = -3 * SD$$

$$upper = 3 * SD$$

Outlier when

$value < lower$ OR $value > upper$

Interquartile Range (IQR)

$$lower = 25percentile - (1.5 * IQR)$$

$$upper = 75percentile + (1.5 * IQR)$$

Outlier when

$value < lower$ OR $value > upper$

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# 4 Addressing outliers

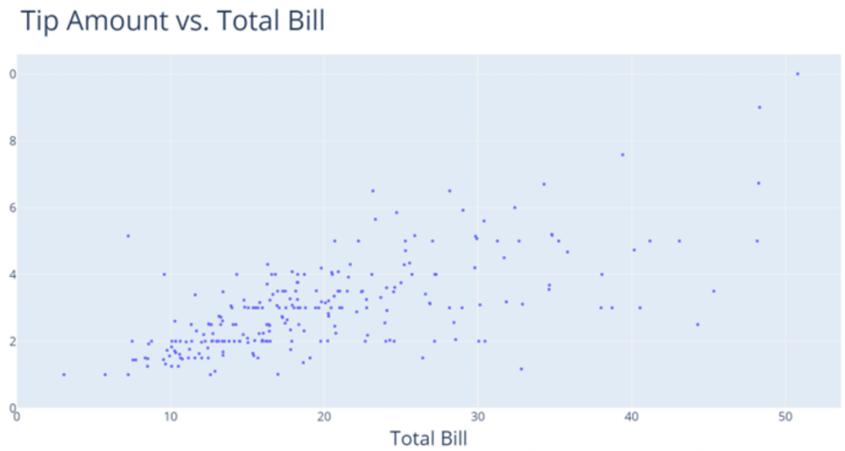
1. Remove observations
2. Imputation

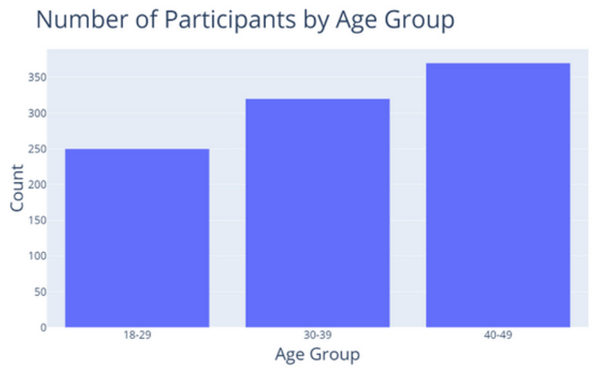
**Winsorizing**

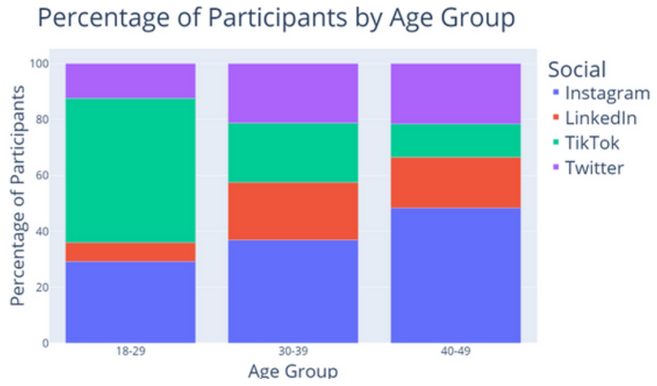
**IF** *value < 5th percentile* **THEN** *value = 5th percentile*

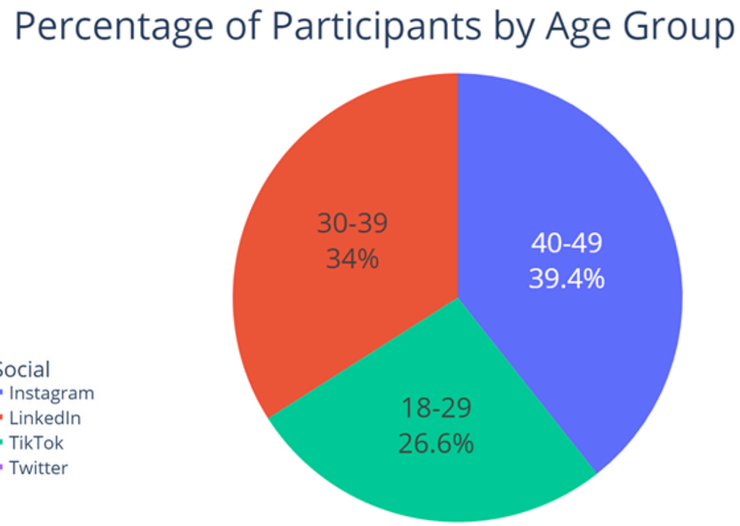
**IF** *value > 95th percentile* **THEN** *value = 95th percentile*

# 5 Examining and quantifying relationships between variables

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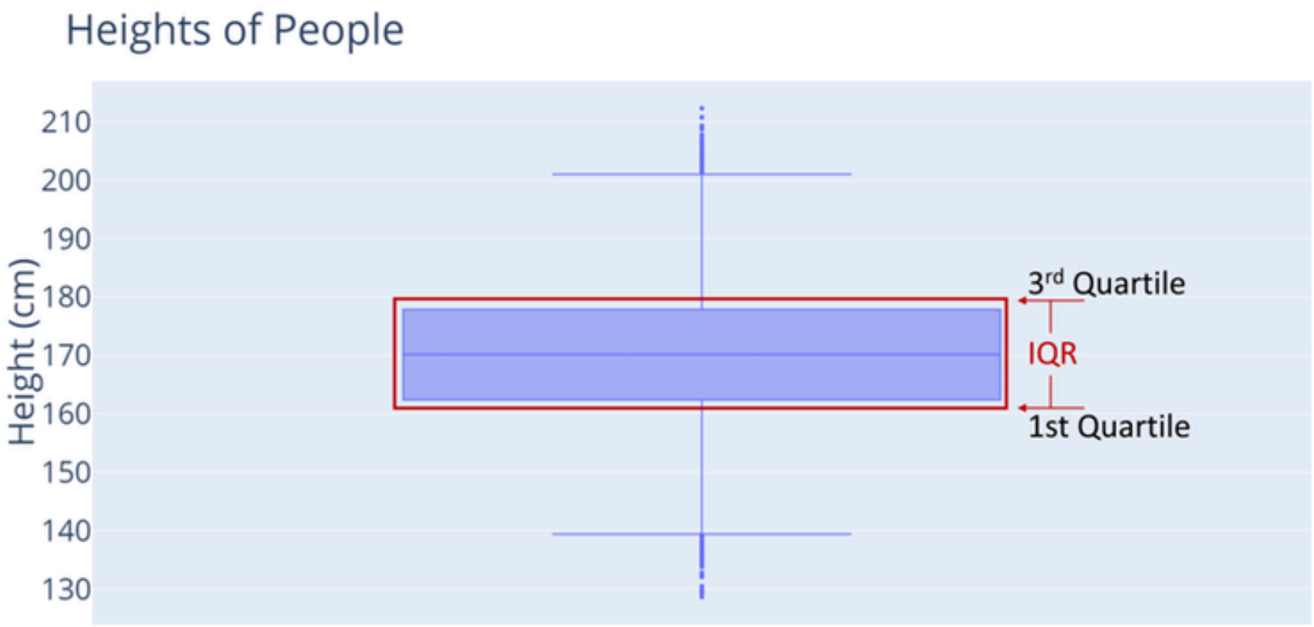
# WHAT ARE BOX PLOTS ?

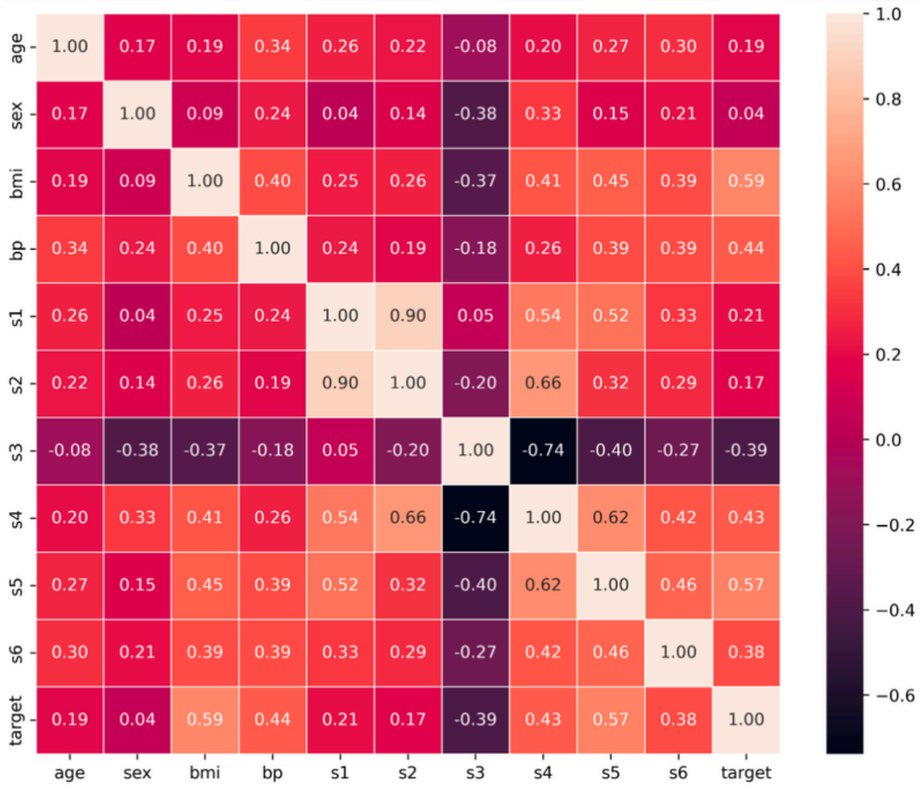


# 5 Examining and quantifying relationships between variables

## EXAMPLE :

## 5 Examining and quantifying relationships between variables

# Correlation Matrix



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6 forming hypothesis

# WHAT IS HYPOTHESIS?

A hypothesis is an educated guess or assumption that we make about our data before testing it.

Example1: Students' Scores

- H0: Male and female students have the same average score.
- H1: Female students score higher than male students.

Example2: Customer Behavior

- H0: Age does not influence purchase frequency.
- H1: Older customers purchase less frequently.

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# THE AI COMP DATA
# “ WAITING FOR YOU “

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# LET'S BE THE WINNERS !

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